

Economic complexity Tests and homeworks

Fall 2025

Test – Sept 23

1. What are the assumptions behind the product space?
2. What are the empirical observations made with the use of product space regarding diversification?
3. What are the implications for policy in less developed countries?

Reading and homework – Sept 23

Read: Hausmann and Hidalgo 2009 PNAS.

Homework: Illustrate the dynamics of complexity ranking in trade, technology and research.

1. Go to the country Complexity Rankings on oec.world and download the country data.
2. Investigate ranking dynamics in all dimensions across 2000-2010-2020. Compare correlations across time but within dimensions. Check cross-correlations across dimensions.
3. /Plus: Merge data with Economic Growth forecast tables <https://oec.world/en/resources/methods/eci-growth-predictions>. Calculate the coefficient of ECI indices using the Solow model of economic growth.
4. Prepare visualizations and discuss the results. Collect figures and text to a pdf (max 2 pages) and save it "givenname_familyname_EC2025_HW1.pdf".
5. Send it to the instructor and TA by email. Deadline: October 6, Monday, 8am. Homework sent after the deadline are not considered.
6. Homeworks covering points 1-2-4-5 can get 2.5 points. Point 3 gets an extra 1.5 point.

Test – Oct 7

1. What does it mean that a country ranks high in economic complexity?
2. What does the method of reflections mean?
3. What do the GDP growth regression models using the economic complexity measures tell us?

Test – Oct 7 (on Oct 21)

1. What does it mean that a country ranks high in economic complexity?
2. Why is the diversity of a country's exports not sufficient on its own to explain economic complexity?
3. What evidence do the authors provide to support the claim that economic complexity is a better predictor of future income growth than initial GDP per capita?

Reading Oct 7

Audretsch and Feldman 1996 AER

Test Oct 14

Summarize the Audretsch-Feldman article in 3 points.

Your 3 points may include:

- Main messages and findings
- Empirical specifications (variables, regressions)
- Explanations or links to previous classes
- Things you liked, did not like or did not understand

Reading and Homework Oct 14

Read: Neffke et al. 2017 *Journal of Economic Behavior and Organization*

Homework: literature search on the urban phenomena.

1. Identify a topic that is connected to spatial concentration of population, or production, or innovation. This can be important for a city, an industry, a country, or the global humanity (eg. automation, inequality, international competition, climate change, populism etc.).
2. Search for journal articles that address this topic with interesting data. Choose one that you would like to show to the class.
3. Prepare 1 pptx slide that summarizes the paper. Include the research question, the data, method and findings. Try to include a visualization that can tell the story. Save the file: "givenname_familyname_EC2025_HW2.pptx".
4. Send it to the instructor and TA by email. Deadline: October 20, Monday, 8am. Homework sent after the deadline are not considered.
5. Homeworks covering 1-2-3-4 can get 2.5 points. 3-5 homeworks can be presented during the Oct 21 class to collect 1 „Activity point” each .

Test Oct 21

<https://forms.gle/9oQiD6CepeK2iS7x5>

Reading October 21

Fleming, L., Charles King III., Juda, A. I (2007). Small worlds of regional innovation. *Organizational Science*, 35(7), 994-1017.

Test Nov 4

Organization Science

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Small Worlds and Regional Innovation

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Reading and homework – Nov 4

Read Tóth et al. Nature Communications 2021

Homework

1. Identify a social or business problem that is related to economic complexity and small-world networks.
2. Prepare a **maximum 1-page** pdf document:
 - a) Describe the importance and relevance of the problem (1 paragraph),
 - b) Describe what we already know how the problem is approached in research, policy, or business. Include citations to papers or websites (1 paragraph),
 - c) Identify a niche that needs better understanding. Formulate a related question (1 paragraph),
 - d) Identify the stakeholders who might be interested and explain how they can benefit from answering your question (1 paragraph),
 - e) Add the references of within-text citations.
3. Save the file: "givenname_familyname_EC2025_HW3.pdf". Send it to the instructor and TA by email. Deadline: November 10, Monday, 8am. Homeworks sent after the deadline are not considered.
4. Homeworks covering 1-2-3 can get 2.5 points. Prepare for a discussion about your homework in the class to collect 1 „Activity point”.

Test Nov 11



ARTICLE

<https://doi.org/10.1038/s41467-021-21465-0>

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Inequality is rising where social network segregation interacts with urban topology

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Reading – Nov 18

Read Tóth et al. Economic Geography 2022

Reading and homework – Nov 25

Read Lengyel et al. Scientific Reports 2020

Homework

1. Identify a limitation/niche that this paper does not cover in the context of spatial innovation diffusion in social networks.
2. Prepare a **maximum 1-page** pdf document:
 - a) Describe this niche (1 paragraph),
 - b) Explain why covering this niche is important for science/society/business (1 paragraph),
 - c) Let me know if you have an idea for methods/data/context to address the niche (1 paragraph),
 - d) What could we learn from this new analysis (1 paragraph),
 - e) Add the references of within-text citations in case you do 2c and 2d as well.
3. Save the file: "givenname_familyname_EC2025_HW4.pdf". Send it to the instructor and TA by email. Deadline: December 1, Monday, 8am. Homeworks sent after the deadline are not considered.
4. Homeworks covering 2a-b can get 2.5 points. Including 2c-d-e can get you additional 1.5 points. Prepare for a discussion about your homework in the class to collect 1 „Activity point”.